QA SYSTEMS & MECHANICS AUDIT

Test Report: Functional Verification & System Defects

Category: Prefab Geometry & Gameplay Subsystems **Scope:** Progression, Inventory, Teleports & Locomotion

1. Verification of New Prefabs & Geometry

geardoor Prefab (Map Spawn / Right Elevation): Detected a rotation and animation state defect — the door swings open in the reverse direction. Additionally, the collider volume is undersized along the vertical axis, allowing the player character to fly directly above the physical door frame and bypass the intended progression barrier.



Figure 1: Intended ground-level approach to the geardoor

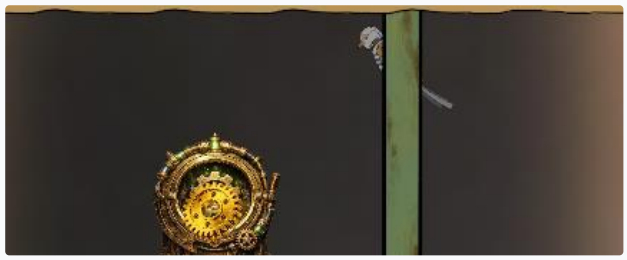


Figure 2: Sequence break – character flying over top collider bounds

Remaining Prefabs (7 New Assets): Functioning properly with correct collision volumes, standard transform scaling, and accurate visual mesh rendering.

2. Systems Functional Test Matrix

Tested System	Status / Behavior	Detected Bugs & Observations (Bug Log)
Checkpoint System	PASS (State Save)	Serialization of inventory items and completed stages operates correctly. However, a logic defect occurs post-death and respawn: previously opened doors reset to their locked resting transform and re-trigger their opening animation when re-approached by the player.
Inventory System	DROP DEFECT (UX)	Pickup triggers and inventory capacity limits perform as designed. A critical bug emerges upon dropping items (gear, crystal) — dropped entities unnaturally inflate in visual scale, hover above the floor plane, and throw console exceptions.
Teleporters & Indicator	LOGIC / UI DEFECT	Teleporters trigger the identical `PlayerCrushDetectorScript.cs` error (`MissingComponentException` at lines 17 and 27) observed with dropped items. The directional guidance arrow exhibits erratic rendering and intermittent visibility.
Dash Mechanic	PASS	Character burst dash behaves reliably across all states. Boundary stress tests demonstrated zero vulnerability to out-of-bounds clipping, floor penetration, or mesh entrapment.

Visual Breakdown of Item Drop Defect



Figure 3: Uncalibrated transform scale upon instantiation



Figure 4: Ground contact failure and levitation artifact

3. Remediation Action Items for Development (To-Do)

- **Geardoor Fix:** Invert the door swing rotation axis in the prefab and expand the vertical bounds of the `BoxCollider2D` to prevent sequence breaking via the upper ceiling gap.
- **Checkpoint Door State Flag:** Persist the door opening boolean state inside the checkpoint serialization struct to prevent world resets after player death.
- **Inventory Drop Normalization:** Normalize the instantiated `transform.scale` of dropped prefabs to match their source assets, and calibrate ground raycasting / physics material settling to eliminate levitation.
- **PlayerCrushDetectorScript Null-Check:** Add explicit null-checks for collider references in `PlayerCrushDetectorScript.cs` (lines 17 & 27) when processing teleporters and dropped entities to silence console exceptions.
- **UI Arrow Lifecycle:** Audit the conditional logic and state triggers controlling the activation and rendering pipeline of the navigation arrow above teleporters.